MANZIL SHRESTHA

Bengaluru, Karnataka, India | +91 7619204739 | shresthamanxil@gmail.com
LinkedIn | GitHub | LeetCode

SUMMARY

Third-year AI & Data Science student building end-to-end machine learning systems, from NLP pipelines to full-stack deployments. Hands-on experience with PyTorch, transformer models (ESM-2), and applied statistical modeling. Seeking ML, data science, or applied AI internship roles where I can ship production-ready solutions.

EDUCATION

CMR Institute of Technology, Bengaluru *2023 - 2027 (Expected)*
Bachelor of Engineering, Artificial Intelligence and Data Science — CGPA: 8.47/10

St. Xavier Higher Secondary School, Jawlakhel, Nepal *2023*
12th Standard (Science) — GPA: 3.2/4.0

TECHNICAL SKILLS

- Languages: Python, C, SQL
- Machine Learning & AI: PyTorch, Scikit-learn, Transformers (ESM-2), NLP, Deep Learning, Model Evaluation
- Data & Databases: Pandas, NumPy, MongoDB, SupaBase, Matplotlib, Excel, Tableau, Power BI
- Web & Tools: HTML, CSS, JavaScript, Git, Linux, XAMPP, Flask, React

PROJECTS

ProteinClassify — Transformer-Based Protein Analysis Platform

- Engineered a full-stack bioinformatics platform that classifies protein sequences across 321 families using transformer embeddings from Meta's ESM-2 model.
- Trained and deployed a custom MLP classifier on extracted embeddings, and integrated ESMFold to generate real-time 3D structure predictions.
- Built batch FASTA upload and PCA-based embedding visualization; delivered with Flask, PyTorch, React, Vite, Tailwind CSS, and the Google Gemini API.

Student Feedback Sentiment Analysis

- Built an end-to-end NLP pipeline that classified 1,000+ student feedback entries into Positive, Neutral, and Negative sentiment categories.
- Achieved 88% classification accuracy with Logistic Regression by applying tokenization, stopword removal, lemmatization, and TF-IDF vectorization; evaluated with precision, recall, and F1-score.
- Automated feedback collection via SMTP and visualized sentiment trends using Pandas and Matplotlib.

SoleTrack — Retail POS & Inventory Management System

- Developed a lightweight POS and inventory system for a retail footwear store with role-based authentication (Admin/Cashier), password hashing, and SKU-level tracking by brand, size, and quantity.
- Built daily sales reports and stock-analytics dashboards to support operational decision-making, using PHP, MySQL/MariaDB, and JavaScript.

EXPERIENCE

The FOOT's Nepal — Design & Requirements Analyst *Aug 2025 - Nov 2025*

- Led requirement analysis for a retail inventory platform (later built as SoleTrack), translating business needs into functional specifications.
- Designed the system architecture and database schema covering products, stock, sales, and users.
- Redesigned the inventory tracking workflow, reducing manual steps and improving day-to-day operational usability.

CERTIFICATIONS & ACHIEVEMENTS

- Top 3 Finish, CMRIT Departmental Project Exhibition (Nov 2025) — Inventory Management System
- Introduction to Cyber Security (Infosys Springboard); Python for Data Science (NPTEL); Fundamentals of DSA, Grade A (CMRIT)
- Google Skill Badges: Prompt Design in Vertex AI; Build Real-World AI Applications with Gemini and Imagen
- Leadership: Promotion Head / Social Media Manager, RARE Club, CMRIT • Participant, Apticode Hackathon 2025